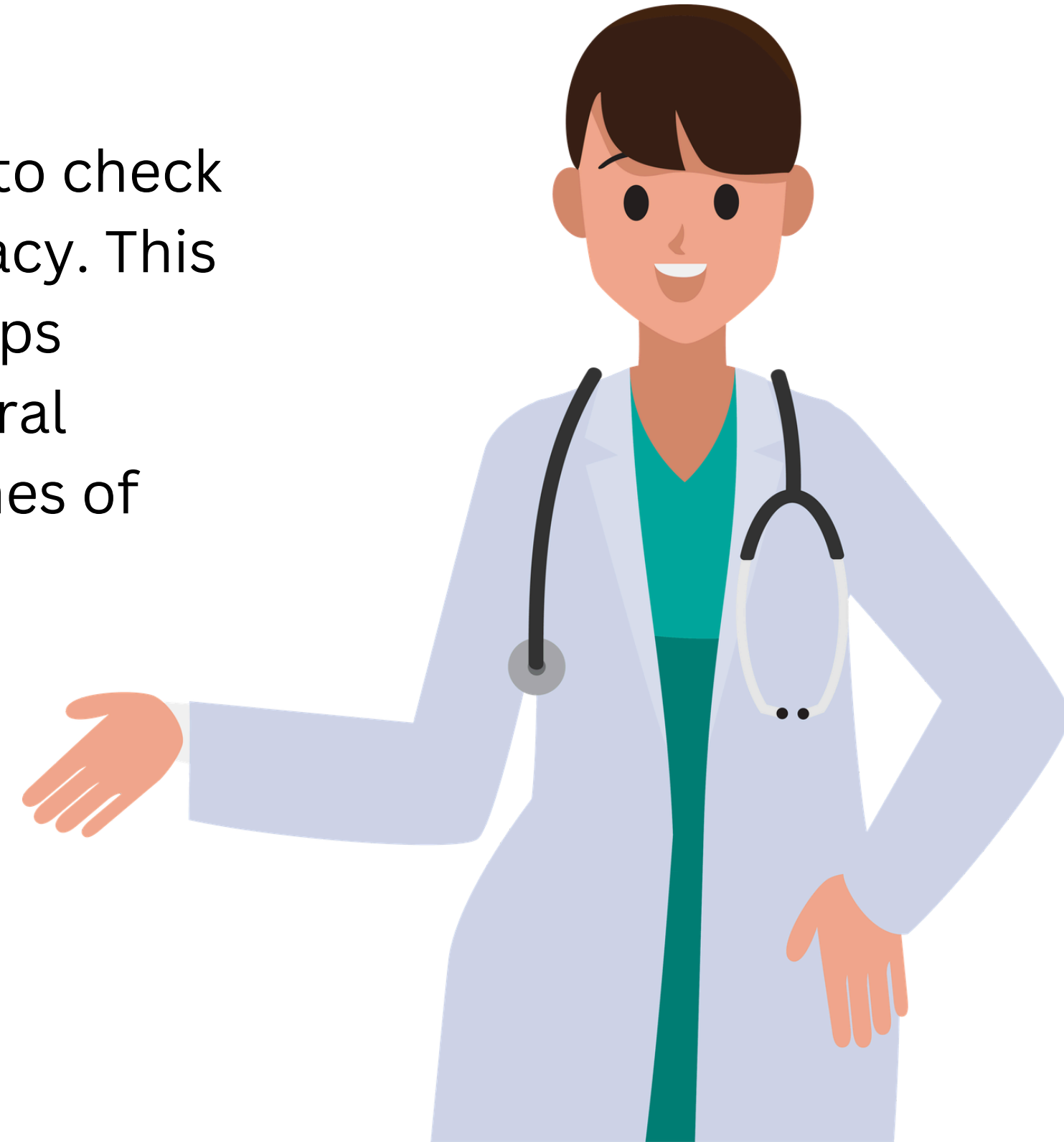




Problem

Create a tool that replicates the CMS automated criteria to check process for network adequacy while ensuring data accuracy. This tool should evaluate county-specialty pairs to identify gaps based on time, distance, and other guidelines set by federal organizations for Medicare, Medicaid, and Commercial lines of business.

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Problem	Stakeholders	Technology	Solution	Metrics & Pitfalls
Understanding the current process <pre> graph LR A[Data Collection] --> B[Network Submission] B --> C[CMS Review Adequacy Check] C --> D[Identify Gaps] D -- Yes --> E[Provider Recruitment] E --> F[Data Validation] F --> G[Update Provider Database] G -- Repeat --> A D -- No --> H[Sell Plan] </pre> Insurers compile a provider list, the Health Service Delivery (HSD) table, with details like specialties and locations. CMS reviews it for adequacy, ensuring enough providers per specialty and accessible locations. If gaps are found, insurers must recruit more providers and keep provider information updated to prevent “ghost networks” of unreachable providers. Problems in the Current Process Inconsistent Data Updates: Providers often submit data irregularly, leading to misalignment with CMS requirements. Manual Recruiting of Providers: Identifying and recruiting providers to fill network gaps is largely manual, time-consuming, and costly. Data Accuracy Issues: Insurers struggle to maintain accurate provider information, due to multiple sources of data leading to ghost networks. High Operational Costs: Managing provider data manually increases costs for large health plans due to labor, time, and repeated recruitment efforts, affecting overall profitability.			53% Patients find incorrect provider info, leading to delays in care and unexpected costs. 31% Medicare Advantage (MA) enrollees had difficulty finding in-network specialists in 2023 49% Provider directory locations had inaccuracies, indicating significant compliance issues in provider data Impact of Solving the Problem Cost Reduction : \$2.76 billion could be saved annually across the healthcare industry by improving provider data accuracy and reducing administrative burdens. (CAQH Report) Better Member Experience: Accurate provider information could reduce member complaints by 30-40%, as they experience fewer issues in finding in-network providers & shorter wait times.(CMS Network Adequacy Requirements) Proactive Network Management: Predictive analytics can help anticipate future provider shortages, potentially reducing the time required for recruitment by 20-30%. (McKinsey & Company) Assumptions Data Sources are Accessible: We assume that CMS databases, Google Maps, and other reliable sources are accessible for verifying provider information. CMS Standards are Stable: We assume that CMS network adequacy guidelines will remain stable, allowing the tool to consistently meet compliance requirements Providers are Willing to Verify Data: We assume that providers will generally respond to verification requests, enabling effective data accuracy maintenance.	

Stakeholders involved

Network Managers

They are directly responsible for compliance with CMS requirements, recruiting new providers and maintaining accurate provider data.

- Tools to monitor and improve network adequacy across regions.
- Efficient methods for recruiting providers and filling gaps.
- Automated systems to ensure data accuracy and avoid outdated information.

Providers

They play a crucial role in meeting the healthcare needs of insured individuals, offering treatments, consultations, and preventive care.

- Providers require a straightforward, low-burden process for updating their information to stay compliant with network and CMS standards.

CMS

sets regulations to ensure networks adequacy requirements, including having a sufficient number of providers per specialty, accessible within a certain time and distance for members

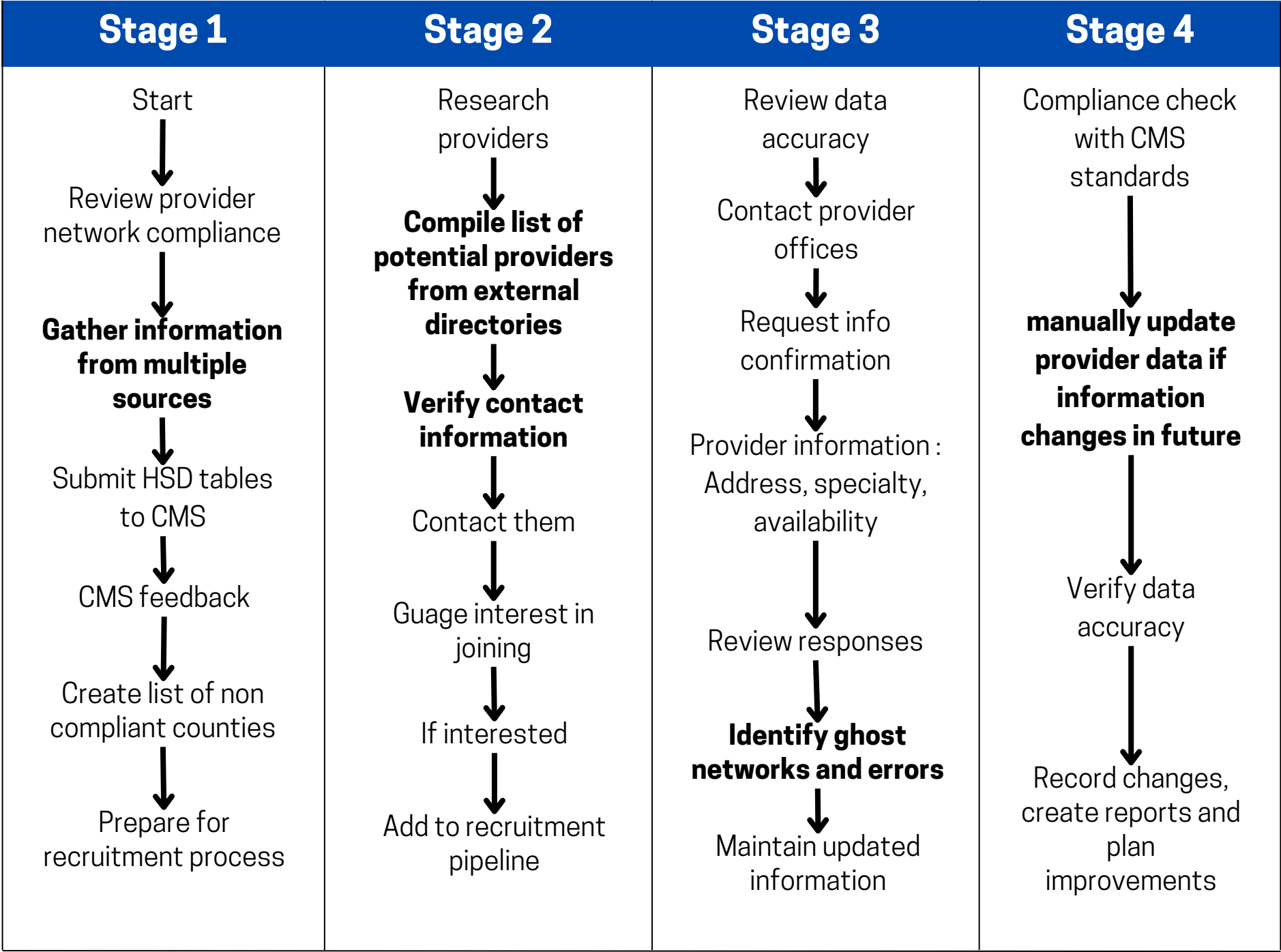
- CMS needs healthcare networks to meet standards for provider availability and accessibility, ensuring members can access care within specific time and distance limits.

Why Network Managers as primary stakeholders?

- Tasked with ensuring that the provider network complies with CMS standards for adequacy and accessibility
- Responsible for filling network gaps by recruiting additional providers, especially in counties or specialties flagged as non-compliant
- Network managed by these managers should ensure accurate, up-to-date provider information and actively prevent ghost networks

User Journey for Network Managers

After some time stage



Pain Points

Gather information from multiple sources

- Data comes from spreadsheets, databases, etc
- Time consuming to compile and standardize information
- Delays compliance process

Compile list of potential providers from external directories

- Requires manual search through multiple directories
- Inconsistent data formats and outdated info
- Difficult to identify reliable candidates

Verify contact information

- Requires manual verification
- Contact information is often outdated or inaccurate
- Increases chance of missed connections with potential providers

Identify ghost networks and errors

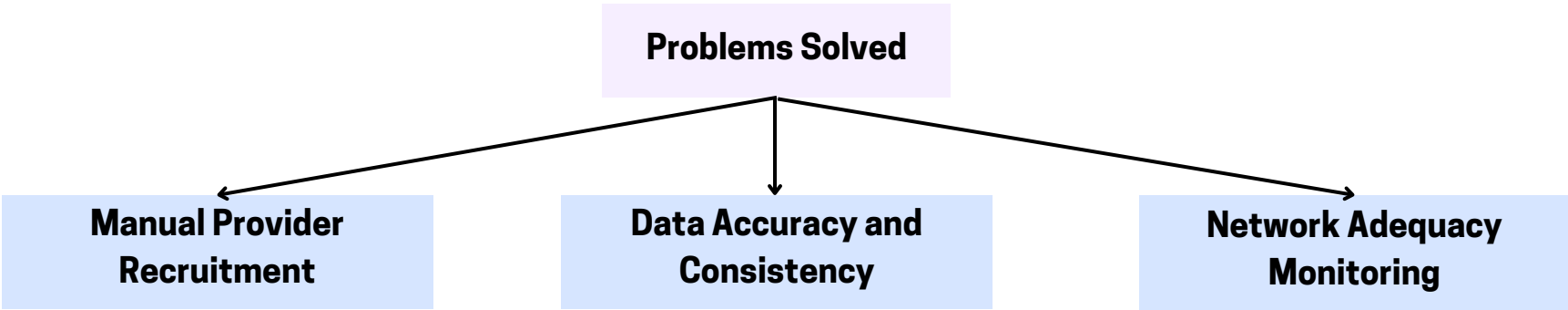
- Hard to detect due to outdated or missing information
- Requires significant time to manually identify and correct errors
- Impacts network reliability and compliance

Manually update provider data if information changes in future

- Updates require repetitive manual entry
- High risk of human error due to ongoing data modifications
- Frequent changes make it hard to keep data current at all times

Introducing NetSure

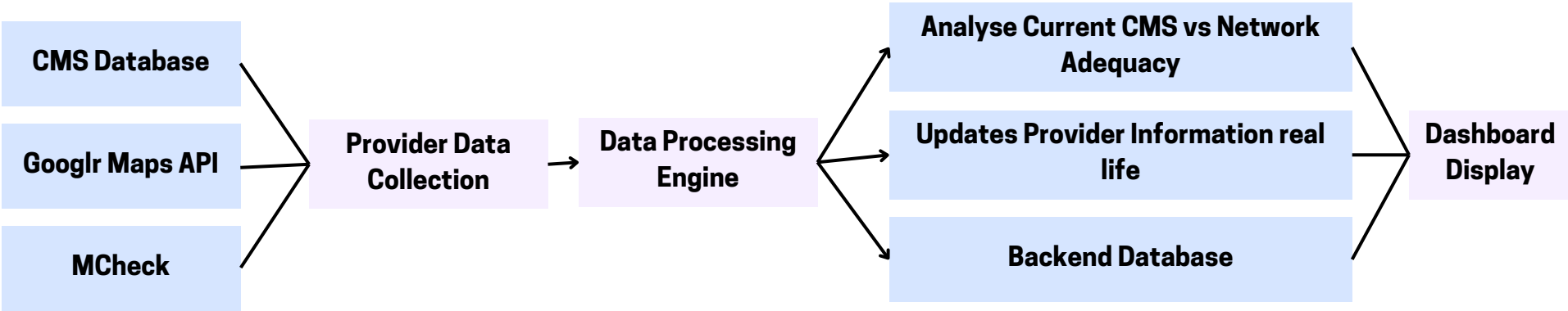
NetSure is a comprehensive network management tool that streamlines CMS compliance, ensures **data accuracy**, and **simplifies provider recruitment**. By automating provider recommendations and outreach, **it reduces recruitment effort**, while **real-time data validation prevents ghost networks** and maintains up-to-date provider details.



Technologies and Integration

Technologies Used and How They’re Leveraged

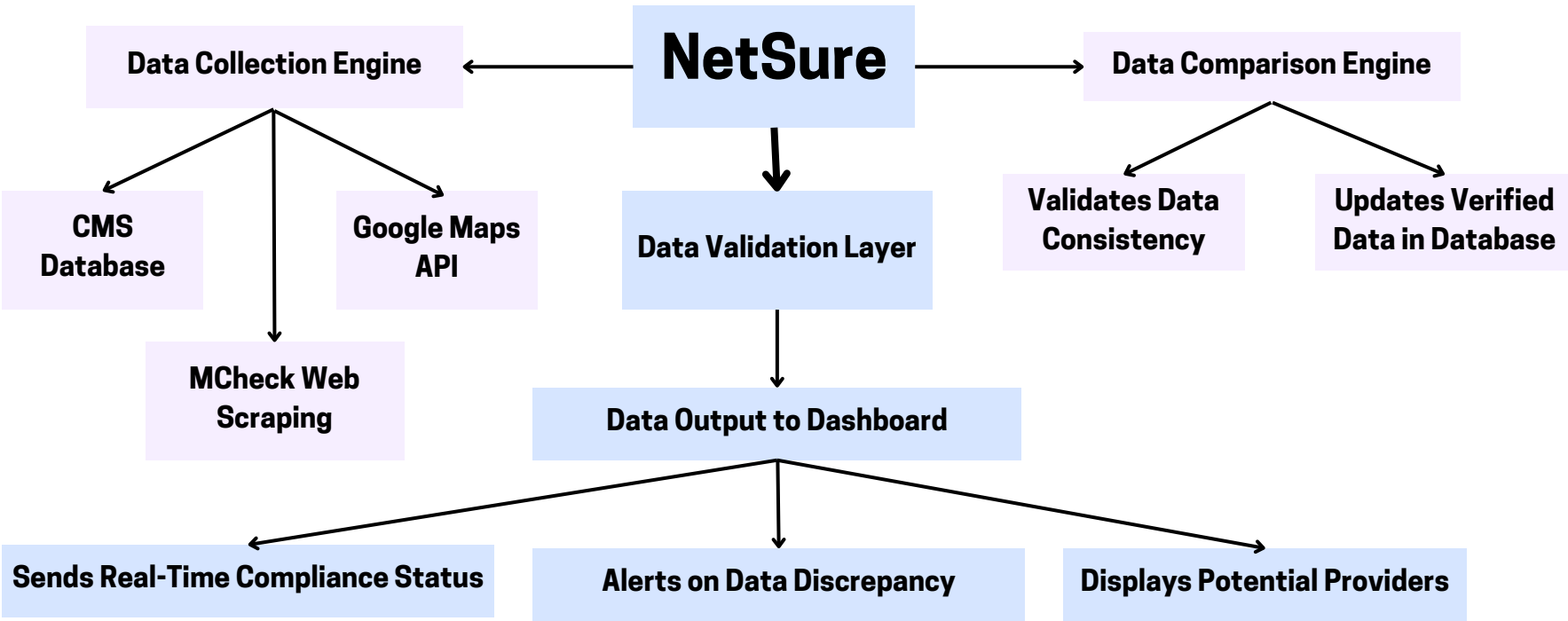
- Google Maps API:** Provides real-time location data for providers, enabling accurate distance and travel time calculations, used for network adequacy checks.
- CMS Database Integration:** Directly accesses CMS provider data and adequacy guidelines, ensuring compliance with federal standards for specialties, time, and distance.
- Automated Web Scraping (e.g., MCheck):** Continually verifies provider data, using web scraping to catch address or specialty changes, maintaining accurate and up-to-date provider directories.
- Automated Communication Tool:** Facilitates bulk outreach to potential providers based on recruitment needs.

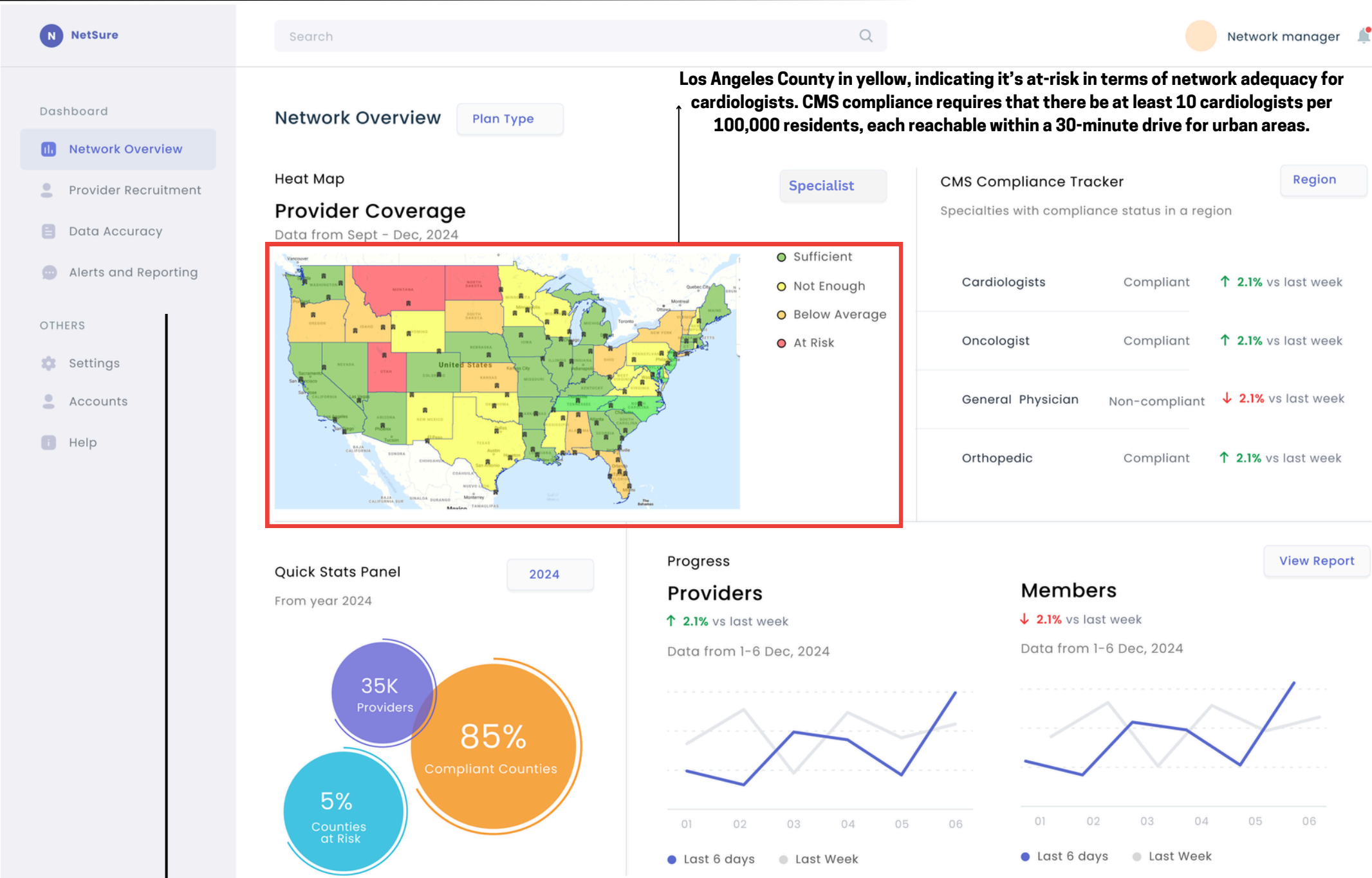


Backend Functionality

How NetSure Works in the Backend

- Data Collection and Validation**
 - Data from CMS and Google Maps: NetSure gathers provider data from CMS and location data via Google Maps.
 - Automated Data Scraping: MCheck validates each provider’s information (like address, contact details), ensuring consistency with external records.
- Data Comparison and Analysis**
 - Network Adequacy Checks: The tool compares provider data against CMS adequacy standards (e.g., sufficient provider count, time/distance compliance) for each county-specialty pair.
 - Gap Identification: Flags counties and specialties not meeting adequacy standards, identifying specific areas needing more providers.
- Provider Recruitment and Data Update**
 - Potential Provider Identification: Analyzes and lists providers close to non-compliant counties, verifying that they accept new patients and meet CMS guidelines.
 - Dashboard Update: Provides a real-time update to the frontend, showing network adequacy, recruitment progress, and data accuracy status.





On main Dashboard screen, it shows 4 main tabs

- Network Overview**
- The Network Overview Tab in NetSure offers a clear snapshot of network health, compliance, and areas needing improvement.
- Provider Recruitment**
- The Provider Recruitment Tab streamlines provider search and recruitment by region and specialty, displaying potential providers who meet CMS criteria.
- Data Accuracy**
- The Data Accuracy Tab ensures provider information is current by automating verification notifications, scoring data quality, and reducing ghost networks and maintaining compliance.
- Alerts and Reports**
- The Alerts & Reports Tab provides real-time alerts for data discrepancies

Network Overview

Provider Coverage

- **Purpose:** Displays a color-coded map of the U.S., indicating provider adequacy by county.
- **Functionality:** This map uses CMS data to compare each county's provider count by specialty against adequacy requirements, highlighting gaps in terms of time, distance, availability criteria.

CMS Compliance Tracker

- **Purpose:** Allows users to choose a specific region and check compliance for various specialties.
- **Functionality:** Details like the count of available cardiologists, primary care physicians, etc., are shown, directly comparing the current provider count with CMS standards.

Quick Stats Panel

- **Purpose:** Summarizes key metrics such as the total number of providers, percentage of compliant counties, etc.
- **Functionality:** Offers a snapshot view of network adequacy status, showing metrics that allow users to monitor overall progress quickly.

Progress Over Time

- **Purpose:** Shows the change in network adequacy from the previous week, including provider recruitment progress and member satisfaction.
- **Functionality:** Weekly data on provider and member trends, enabling users to assess if the network is progressing or if more focus is needed on certain areas.

Provider Recruitment

Potential Providers

- Purpose :**
- Enables network managers to search for providers by region and specialty, displaying all available providers who meet adequacy criteria.
 - Allows users to send automated connection emails to selected providers, inviting them to join the network.
- Functionality :**
- Users specify a region and specialty (e.g., cardiologists in Los Angeles), and the tool displays a list of potential providers in that area who meet CMS requirements for time, distance, and other guidelines.
 - Shows detailed information for each provider, such as name, specialty, location, contact details, and status of accepting new patients, with filters for those open to new patients.
 - Users select providers from the list to initiate a pre-written recruitment email, with tracking and follow-up reminders if providers do not respond.

N

NetSure

Search

Q

Network manager

Dashboard

Network Overview

Provider Recruitment

Data Accuracy

Alerts and Reporting

OTHERS

Settings

Accounts

Help

Provider Recruitment

Automated Outreach

Potential Providers

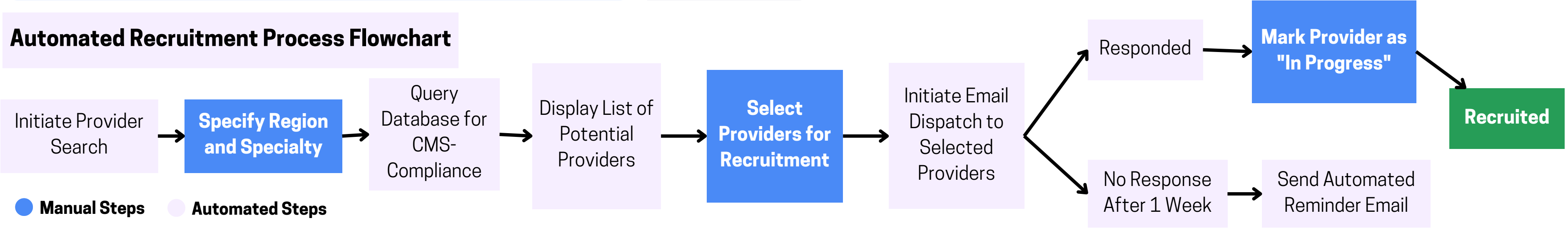
Data from Sept - Dec, 2024

Region

Specialist

Name	Contact	Address	Accepting New Patients	Mail
Dr. Alex Benjamin	9829xxxx12	51, Jane street, New York	Yes 10 Patients	Send
Dr. Emma Watson	9829xxxx12	52, Jane street, New York	Yes 5 Patients	Send
Dr. Chris Algo	9829xxxx12	53, Jane street, New York	NO	Not Available
Dr. Stone Guy	9829xxxx12	54, Jane street, New York	Yes 5 Patients	Send
Dr. Harry Potter	9829xxxx12	55, Jane street, New York	Yes 5 Patients	Send
Dr. Ron Won	9829xxxx12	56, Jane street, New York	NO	Not Available
Dr. Stephan Strange	9829xxxx12	57, Jane street, New York	Yes 5 Patients	Send

Automated Recruitment Process Flowchart



NetSure

Search

Network manager

Dashboard

Network Overview

Provider Recruitment

Data Accuracy

Alerts and Reporting

OTHERS

Settings

Accounts

Help

Data Accuracy

Automated Data Maintenance

Registered Providers

Data from Sept - Dec, 2024



Dr. Harry Potter

Outdated

Directory Listed
Address : 123, Jane Street, Queens road, United States of America
Working Hours : 9:00 AM TO 3:00 PM
Accepting Patients : Yes
Contact Info : 9829xxxxx
Email Address : Harryp@gmail.com

Last Update Due : 25th Mar, 2024

Notification Status :

Sent

 (Automatically 1 Week ago)

Call to confirm

Data Quality Scores

Quality of each provider's data

Low

Name	Score	Review	Take Action
 Dr. Alex Benjamin	10	Very Poor data response	Action
 Dr. Jon Doe	24	Poor data response	Action
 Dr. Stephan	31	Poor response	Action

Data Verification Status

Tags providers with verified or not verified

Dr. Alex Benjamin

Verified

Dr. Harry Potter

Outdated

Dr. Snape

Verified

Dr. Jon Won

Verified

Dr. Stephan

Review Pending

Dr. ABC

Verified

Dr. DEF

Verified

Dr. GHI

Review Pending

Dr. JKL

Review Pending

Dr. MNO

Outdated

Dr. PQRS

Verified

Dr. XYZ

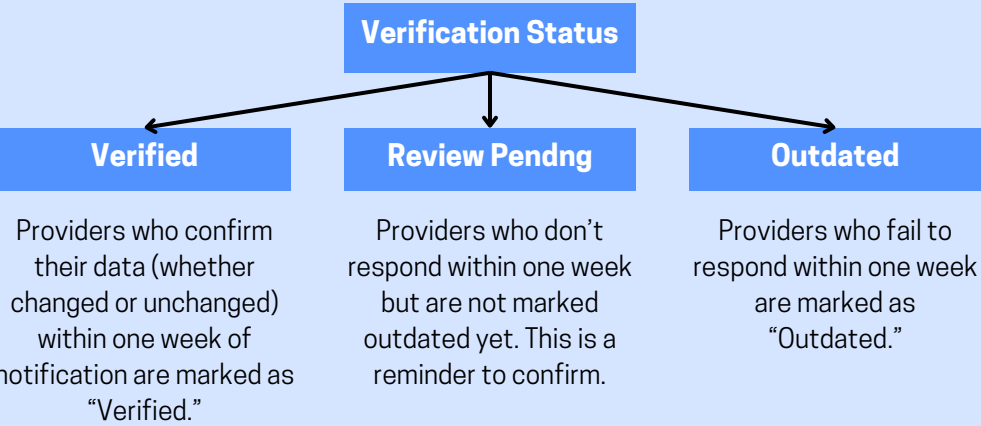
Outdated

Network managers see Dr. Harry’s status as “Outdated” and can click a “Call” button to automatically initiate a reminder call. If confirmed, the status updates to “Verified.”

Data Accuracy

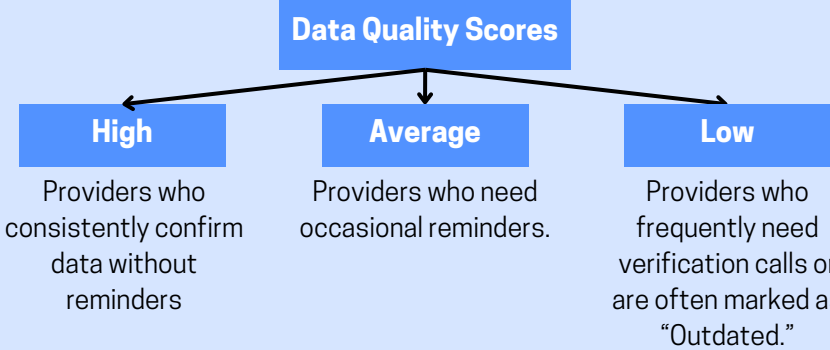
Data Verification Status

- **Purpose:** Tracks the verification status of each provider’s information in the network.
- **Functionality:** Providers receive a verification notification at set intervals (e.g., every 6 months) to confirm if their data (address, specialty, contact info) is still accurate.



Data Quality Scores

- **Purpose:** Scores providers based on factors like response time to data verification notifications, frequency of reminders needed, and the accuracy of submitted data.
- **Functionality:** For providers with low scores, options include sending another notification, setting up a call, or removing them from the network if their data accuracy doesn’t improve.

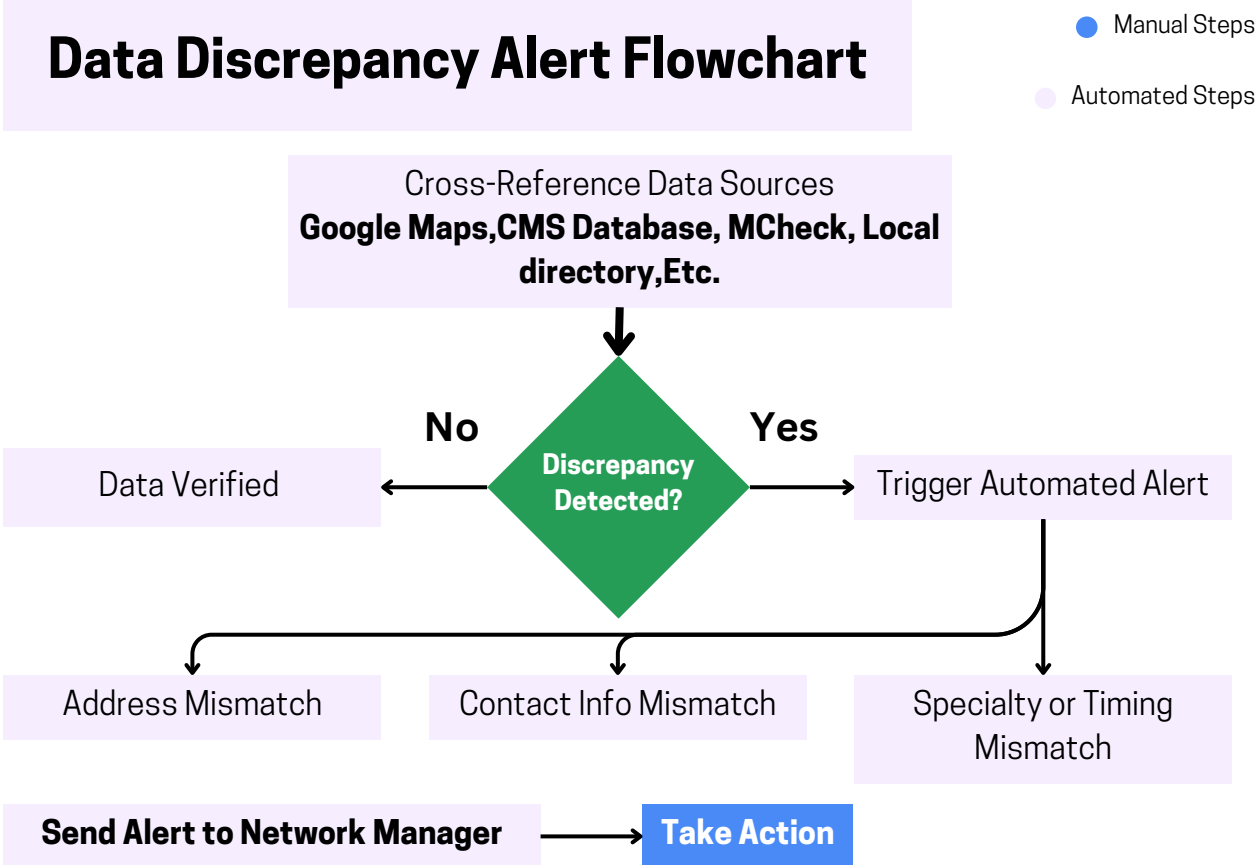


Alerts and Reporting

Data Discrepancy

- **Purpose:** Detects mismatches in provider data, such as address or contact info, by cross-referencing multiple sources.
- **Functionality:** Integrates with Google Maps, CMS databases, and local directories to cross-verify provider addresses, specialties, contact information, and office hours. If discrepancies are detected (e.g., a provider’s address on Google Maps differs from the directory entry), an automated alert is triggered for immediate action.

Data Discrepancy Alert Flowchart



NetSure

Search

Network manager

Dashboard

Network Overview

Provider Recruitment

Data Accuracy

Alerts and Reporting

OTHERS

Settings

Accounts

Help

Alerts and Reporting

Real-time alerts

Data Discrepancy

Data from Sept - Dec, 2024

Total Discrepancy : 10

Dr. Smith

Address Mismatch Detected

MCheck : 123, main street, New York
Google Maps : 51, Queens road, New York
Local Directory : 51, Queens road, New York

Connect

Dr. Jon

Contact Mismatch Detected

MCheck : (+100) 89xxxxxxx
Google Maps : (+100) 91xxxxxxx
Local Directory : (+100) 23xxxxxxx

Connect

Dr. Harry

Timings Mismatch Detected

MCheck : Mon-Fri 9:00 am - 5:00 pm
Google Maps : Tue, Wed 9:00 am - 9:00 pm
Local Directory : Mon-Fri 9:00 a - 5:00 pm

Connect

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Metrics to look	Provider Acquisition Rate	Data Accuracy Rate	Compliance Rate	Response Rate to Verification Notifications	Member Satisfaction
	Number of new providers added to the network via NetSure who meet CMS adequacy requirements	Percentage of verified provider profiles with up-to-date information on NetSure	Percentage of counties and specialties that meet CMS adequacy standards	Percentage of providers responding to data verification notifications within the required timeframe.	Member feedback scores related to provider accessibility and network adequacy.

Pitfalls and Solutions

Pitfalls	Solution
Lack of Incentives for Providers to Maintain Accurate Data	Implication of a rewards system where providers with consistently accurate and updated data receive benefits, such as increased visibility in the network or priority ranking for new patient assignments
Delayed Response from Providers During Recruitment	Automated follow-up emails or calls with personalized messages to encourage timely responses.
Difficulty Identifying and Prioritizing Critical Data Discrepancies	Implement a tiered alert system that categorizes discrepancies by severity, such as critical (e.g., address mismatches) and minor (e.g., timing updates)

Future Enhancements

- **Member Feedback Integration** involves collecting direct feedback from members about their experiences with provider accessibility and network adequacy. By surveying members on factors like ease of finding in-network providers, travel time, appointment availability, and overall satisfaction, NetSure can gain valuable insights into areas where the network meets expectations and where improvements are needed. This approach helps identify gaps in service quality, ensuring that network adjustments are not only CMS-compliant but also aligned with member needs and satisfaction
- **Provider Performance Tracking** involves establishing metrics to evaluate each provider’s effectiveness and impact within the network, focusing on factors such as patient feedback, appointment availability, and accessibility. This performance data enables network managers to make informed decisions on retaining high-performing providers, addressing issues with lower-performing ones, and identifying ideal candidates for recruitment. Ultimately, this ensures that the network not only meets regulatory standards but also maintains a high level of care quality for members.